CASE STUDY: LARAVEL STUDENT PORTALS (MTQ UNY NATIONAL COMPETITION)

Role:	Backend Web Developer
Framework:	Laravel 7 (PHP), MySQL, Bootstrap 4
Status:	Archived / Completed (2021)
Video Demo / Walkthrough:	https://www.youtube.com/watch?v=dCGDNHVLXLk

1. Project Overview

The MTQ UNY Portal was designed to manage a national-level student competition. Hosting universities from all across Indonesia, the system handled the end-to-end registration, verification, and digital submission processes for 12 different competition categories (ranging from Al-Quran recitation to Computer Application Design). Previously hosted on *mtqm.uny.ac.id* (and later migrated to *lomba.uny.ac.id*), it served as a robust, centralized hub for national participants and internal verifiers.

2. Key Engineering Challenges & Solutions

A. Dynamic Multi-Category Submissions

The Challenge: The competition featured 12 highly distinct categories. Some categories required teams to upload PDF proposals, others required pasting YouTube video URLs for preliminary rounds, and some were entirely live-judged without prior submissions. Hardcoding the logic for each category would make the database rigid and difficult to scale for future events.

The Solution: Using Laravel's Eloquent ORM, I designed a flexible, relational database architecture. I abstracted the submission requirements into a polymorphic or highly relational structure, allowing admins to dynamically dictate whether a specific division

required a file upload or a URL. This kept the controllers lean and the codebase highly maintainable.

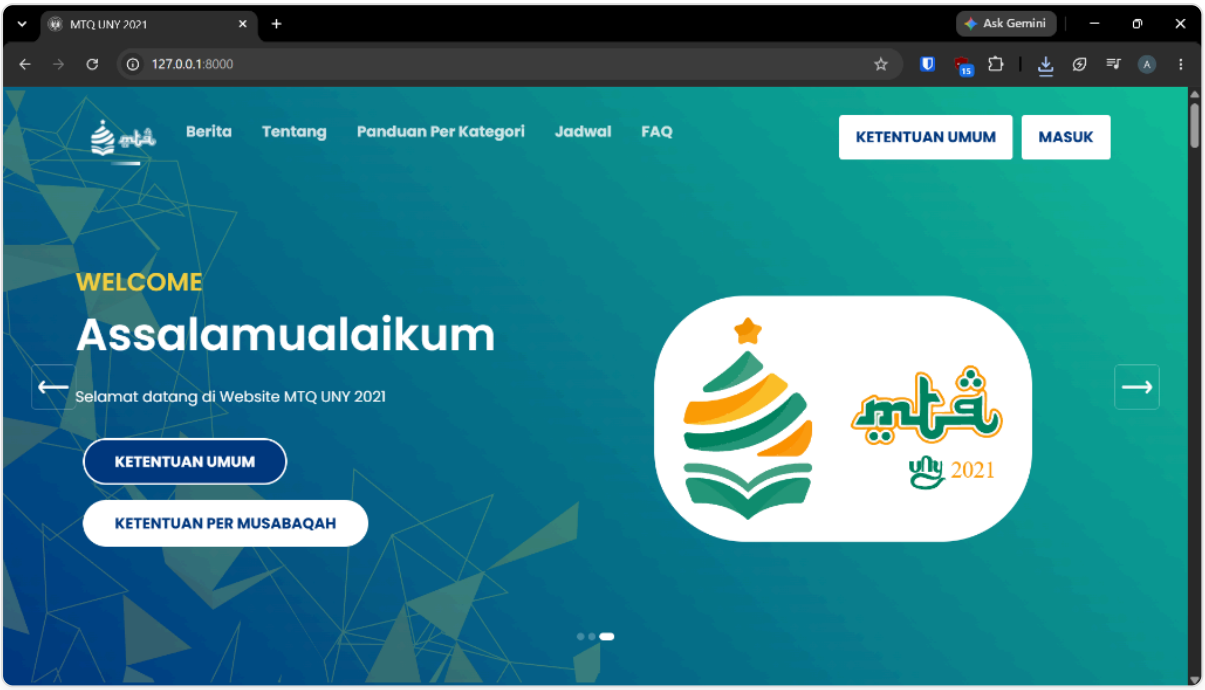
B. National-Scale Participant Verification

The Challenge: With participants arriving from various universities (Perguruan Tinggi) across the nation, validating student identities and team structures was a massive administrative bottleneck. The system needed real-time tracking of which universities were verified.

The Solution: I integrated DataTables with server-side processing to handle the large datasets of universities and teams without freezing the browser. The dashboard provided admins with a clear, fast UI to approve teams and publish live statistics of verified institutions directly to the homepage, increasing transparency for participants.

3. System Interface & User Flow

(Please refer to the embedded YouTube tutorial link above for a complete walkthrough of the internal dashboard and registration flow).



The Landing Page: MTQ UNY National Student Portal.